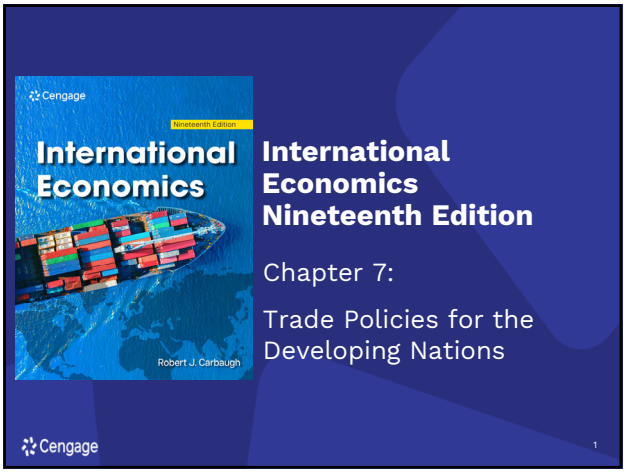


1

Chapter Objectives

After completing this chapter, you should be able to:

7.1 List and characterize the trade problems faced by developing nations, as well as methods used to address them.

7.2 Describe the role and function of the Organization of Petroleum Exporting Countries (OPEC).

7.3 Describe the role and function of international aid agencies including the World Bank and International Monetary Fund (IMF).

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2

Chapter Objectives

7.4 Compare different economic growth strategies including import substitution and export-led

7.5 Summarize the economic growth performance of selected developing nations, including China and India.

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Developing Nations Characteristics

- Nations are divided by real income into advanced (e.g., North America, Western Europe) and developing (e.g., Africa, Asia)
- Advanced Nations: high GDP per capita, long life expectancy, high literacy
- Developing Nations: most of global population, poorer, face trade system challenges

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Developing Nations Characteristics

- Trade Characteristics:
  - Major exports to and imports from advanced nations; limited intra-developing trade
  - Emphasis is on primary products (e.g., agriculture, fuels); labor-intensive manufactures (e.g., textiles)

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Developing Nations Characteristics

- In the last 30 years, nations like China, India increased manufactured exports, reducing poverty
- Drivers:
  - Education
  - Capital investment
  - Transport improvements
  - Trade liberalization

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Developing Nations Characteristics

- Some developing nations have not integrated strongly into the global industrial economy
- Exports consist of a narrow range of primary products

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Developing Nations Characteristics

- Challenges:
  - Poor infrastructure
  - Inadequate education
  - Rampant corruption
  - High trade barriers
  - Transport costs

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Table 7.1 Basic Economic and Social Indicators for Selected Nations, 2022

Country	Gross Domestic Product Per Capita ($)*	Life Expectancy (years)	Adult Literacy (percent)	
			Male	Female
Switzerland	83,598	84	99	99
United States	76,399	79	99	99
Japan	45,573	84	99	99
Chile	30,209	80	96	96
Mexico	21,512	75	96	95
Algeria	13,210	76	87	75
Guinea	3,187	61	44	22
Burundi	836	61	75	61

*Converted into international dollars using purchasing power parity rates. An international dollar has the same purchasing power as a U.S. dollar has in the United States.
Source: From the World Bank at http://www.worldbank.org/data

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Tensions Between Developing Nations and Advanced Nations

- Scholars and policy makers advocate leveraging trade for development; many nations opened markets
- Advanced nations protect against developing imports, creating tension

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Tensions Between Developing Nations and Advanced Nations

- Economic Ladder
  - Developing nations at bottom (textiles) must displace advanced producers to climb
- Protectionism
  - Shields low-income workers in advanced nations, complicating global equity

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Tensions Between Developing Nations and Advanced Nations

- Structural weaknesses in developing economies persist due to nonexistent or inadequate institutions and policies in:
  - Law and order
  - Sustainable macroeconomic management
  - Public services

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Unstable Export Markets

- Advanced nations see trade benefits; developing nations question equity
- Protectionism hinders industrialization; seek new trade order

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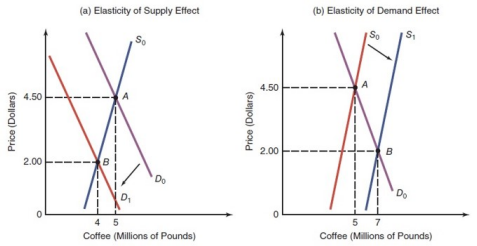
Unstable Export Markets

- Key Issues:
  - Unstable export markets
  - Worsening terms of trade
  - Limited market access
  - Exports often one/few products

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Figure 7.1 Export Price Instability for a Developing Nation



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Worsening Terms of Trade

- Developing nations complain that commodity terms of trade have deteriorated
  - Export prices fall relative to imports, favoring advanced nations
- Causes:
  - Advanced monopoly power vs. competitive primary markets
  - Demand shift to manufacturers

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Limited Market Access

- Global protectionism is a hindrance to developing nations' market access in agriculture, and labor-intensive manufactured goods
  - Tariffs: Higher on developing vs. advanced imports; tariff escalation limits value-added exports
  - Quotas ended (2005), but high tariffs, antidumping duties persist
  - Developing nations urged to lower own high tariffs

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Table 7.2 Tariffs of Selected Developing Nations and Advanced Nations, All Products, 2022

Country Average	Applied Tariff Rate, Percent*
India	39.6
Sudan	30.7
Barbados	29.0
Algeria	23.7
Ethiopia	22.5
Ecuador	17.4
Japan	13.4
European Union	11.4
United States	5.1

*Average applied tariff rates are duties that are actually charged on imports. These can be below the bound rate, which is the maximum tariff rate for a given product that a country has committed not to exceed. World Trade Organization members can apply tariffs at any level up to their bound level.

Source: From the World Trade Organization, *World Tariff Profiles*, 2022.

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Agricultural Export Subsidies of Advanced Nations

- Subsidies
  - Displace developing exports
  - Dump surpluses
  - Depress prices
- Examples: West African rice, cotton farmers vs. U.S./EU subsidies

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Bangladesh's Sweatshop Reputation

- 1974–2005: World trade in clothing was governed by the Multifiber Arrangement (MFA), which imposed quotas on the amount developing countries could export to developed countries
- Bangladesh's labor was relatively cheap
  - Low-wage clothing boom post-MFA (2005); 5,500 factories by 2013
  - Unsafe factories led to 2013 collapse (1,100+ deaths)
- Response: European firms fund safety; Walmart blacklists 250 factories; standards vary

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Production and Export Controls

- International Commodity Agreements (ICAs) attempt to stabilize revenues of primary producers
- ICAs relied on production and export controls, buffer stocks, multilateral contracts
  - Six agreements (e.g., coffee, tin) aimed to stabilize prices, collapsed by 1980s–90s
  - Evolved into "study groups" agreements to collect and disseminate market information, market promotion, conduct research and development

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Production and Export Controls

- If an ICA accounts for a large share of total world output (or exports) of a commodity, its members may agree on production and export controls to stabilize export revenues
  - Quotas set to maintain target prices (e.g., tin)
  - Affect price; influence world supply; are based on target price

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Production and Export Controls

- Obstacles:
  - Allocation disputes
  - New entrants
  - Cheating hard to enforce

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Buffer Stocks

- Establishment of a producers' association (or international agency)
- Buffer stock consists of supplies financed and held by producers' association
- Buffer stock manager buys low, sells high

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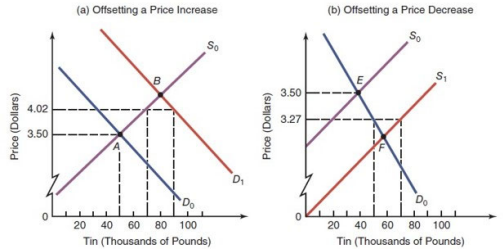
Buffer Stocks

Advantages	Disadvantages
• Economic efficiency	• Target price setting difficult
• Moderate price inflation of industrialized nations	• High costs of holding stocks
	• Poor decisions

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Figure 7.2 Buffer Stock: Price Ceiling and Price Support



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Multilateral Contracts

- Multilateral contracts are another method of stabilizing commodity prices
  - Minimum price at which importers will purchase guaranteed quantities
  - Maximum price at which producing nations will sell guaranteed amounts to importers
- Advantage
  - Less distortion of market mechanism and allocation of resources
- Limited stability due to easy exit/entry

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Does the Fair-Trade Movement Help Poor Coffee Farmers?

- Objective of fair-trade coffee movement is to increase income to farmers by setting up system for them to bypass middlemen & sell directly to roasters & retailers
  - Nicaraguan farmers earn $1.26 vs. $0.40 per pound via direct sales
  - 500,000 farmers joined; strong in Europe, less in U.S.
  - Critique: Retailers gain most; consumer awareness low

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What is the OPEC Oil Cartel?

- Cartels form among exporting nations to increase price and realize “monopoly” profits
- The Organization of Petroleum Exporting Countries (OPEC) is a group of oil-rich nations that sells oil on the world market

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What is the OPEC Oil Cartel?

- OPEC:
  - Controls 40 percent of oil output; expanded to OPEC+ (2016) with Russia
  - Raised prices from $3 to $36 (1973–1981); fell to $11 (1986)
  - Shifted from pre-OPEC competition to quota-based control

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Table 7.3 The Ten Largest Oil Producers and Share of World Oil Production in 2022

Country	Million Barrels Per Day	Percent of World Total
United States	20.21	20
Saudi Arabia	12.14	12
Russia	10.94	11
Canada	5.70	6
China	5.12	5
Iraq	4.55	5
United Arab Emirates	4.24	4
Brazil	3.77	4
Iran	3.66	4
Kuwait	3.02	3
Total, Top Ten	73.36	73
World Total	99.89	

Source: Data drawn from U.S. Energy Administration, International Energy Statistics, Total Oil (Petroleum and other Liquids) Production as of May 1, 2023, available at https://www.eia.gov.

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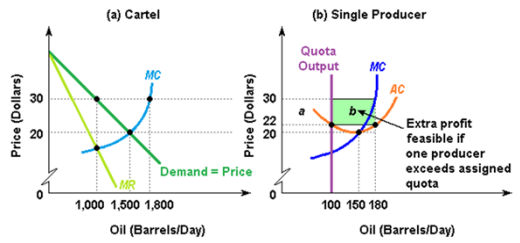
Maximizing Cartel Profits

- Obstacles to cartel profit maximization:
  - Many sellers
  - Cost/demand differences
  - Potential competition
  - Economic downturn
  - Substitute goods

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Figure 7.3 Maximizing OPEC Profits



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OPEC as a Cartel

- OPEC is not fully a cartel; quotas often exceeded since 1983
- Saudi role is to use spare capacity to influence prices

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OPEC as a Cartel

- Countermeasures:
  - Raise the fuel economy standards
  - Increase federal excise tax on gasoline
  - U.S. boosts supply (fracking, drill on federal land)
  - Diversify imports
  - Explore alternatives (biofuels and wind power)

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The World Bank

- Developing nations have sought aid to offset trade system flaws
- Institutions and policies include World Bank, IMF, GSP

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The World Bank

- World Bank:
  - Loans/grants since 1944 for poverty reduction (189 members)
  - Five entities aid projects (e.g., AIDS in Guinea, schools in Bangladesh)
- Issues: Corruption diverts funds; role wanes as China, others fund

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International Monetary Fund

- International Monetary Fund (IMF)
  - Aids deficits with loans from quotas, lines of credit (189 nations)
  - Surplus nations channel funds to nations with temporary deficits
- Conditionality: austerity required, facing resistance
- Critique: moral hazard, recession risk from tight policies

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Generalized System of Preferences (GSP)

- Major advanced nations temporarily reduce tariffs on designated manufactured imports from developing nations below levels applied to imports from other advanced nations
  - Duty-free access for 3,500 items from 119 nations (U.S. since 1976)
  - Excludes textiles; expired 2020; eroded by nontariff measures
  - Reauthorization hinges on eligibility criteria

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Does Aid Promote Growth of Developing Nations?

- Growth-oriented aid effective, not universal
- Critics: aid fosters corruption; favors wealthy in poor nations; is often squandered; fails in nations like Haiti
- Proponents: boosts growth in South Korea, others; infrastructure aid yields 13 percent return

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What are Economic Growth Strategies: Import Substitution versus Export-Led Growth?

- Developing nations have pursued two competing strategies for industrialization:
  - Inward-looking strategy (import substitution)
  - Outward-looking strategy (export-led growth)

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Import Substitution

- Inward-oriented (import substitution)
  - Protect domestic industries (1950s–60s: Argentina, Brazil)
  - Pros: low risk, local jobs
  - Cons: inefficiency, small markets, corruption
  - Freer trade policies grew faster by 1980s

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Export-Led Growth

- Outward-looking strategy (export-led growth)
  - Promote comparative advantage exports (post-1980s)

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Export-Led Growth

- Advantages
  - Encourages industries in which developing nations are likely to have a comparative advantage such as labor-intensive manufactured goods
  - Allows domestic manufacturers greater scope for exploiting economies of scale
  - Imposes a competitive discipline on domestic firms that forces them to increase efficiency

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Is Economic Growth Good for Everyone?

- Critics: Trade growth harms poor via instability, inequality
- Evidence that growth does aid the lowest-income individuals in developing countries
- Liberal economic policies
  - Opens markets and fosters monetary and fiscal stability
  - Raises incomes of poor and everyone else in society proportionately

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Can All Developing Nations Achieve Export-Led Growth?

- Simultaneous exports may lower prices, spur protectionism
  - Developing nations' exports small (5 percent world output)
  - Intra-trade, specialization viable

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Flying-Geese Pattern of Growth

- East Asian nations are highly diverse in natural resources, populations, cultures, and economic policies
  - High investment, education; open trade, competitive markets
  - Shifted from import substitution to export push (1950s–60s)
- Issues: Pollution, U.S. protectionism

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Flying-Geese Pattern of Growth

- Nations gradually move up in technological development
  - by following pattern of nations ahead of them in development process (e.g., Japan to Taiwan to Thailand)

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Flying-Geese Pattern of Growth

- Drivers:
  - Market forces
  - Export platforms that create incentives for labor-intensive exports
  - Example: Bonded warehouses, free-trade zones, joint ventures, strategic alliances with multinational enterprises

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Table 7.6 East Asian Economies' Annual Percentage Growth Rates of Real Gross Domestic Product, 2013–2022

Country	2013	2016	2019	2022
China	7.8	6.9	6.0	3.0
Cambodia	7.4	6.9	–3.1	2.1
Philippines	6.8	7.1	6.1	7.6
Indonesia	5.6	5.0	5.0	5.3
Thailand	2.7	3.4	2.2	2.6
Vietnam	5.6	6.7	7.2	8.0
Malaysia	4.7	4.5	4.4	8.7

Source: Data drawn from the World Bank, World Data Bank, World Development Indicators, available at www.databank.worldbank.org. See also Central Intelligence Agency, World Fact Book, available at www.cia.gov

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Industrial Policy

- Before the 1700s: China had one of the world's largest and most advanced economies
- 1800s and early 1900s: economy stagnated and even declined in absolute terms
- Following World War II: revolutionary communist movement captured control
- 1970s: China began marketizing its economy
- 2001: Entered the WTO

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Industrial Policy

- Industrial policy to become a world leader in high-technology manufacturing
- Measures include:
  - Governmentally established output targets for private and public firms
  - Provision of governmental subsidies to Chinese firms

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Industrial Policy

- Measures include:
  - Encouragement to obtain access to advanced technology by investing in foreign firms producing semiconductors
  - Mobilizing state-backed, Chinese firms such as ZTE and Huawei
  - Adopting forced technology agreements with overseas firms

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Challenges and Concerns for China's Economy

- Rising labor costs
- Development of infrastructure
- Reliance on investment spending
- Environmental future
- Convertibility of the yuan
- Concerns about intellectual property rights
- Holdings of U.S. securities
- Belt and Road initiative (BRI)
- New productive forces

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Forced Technology Transfer and China

- Technology transfer is the process of disseminating methods of production from the originator to a secondary user
- Results in the sharing of methods of production to other users who can then further develop and exploit the technology into new products
- Foreign companies often maintain that this is a customary practice in China
- Example: GE, Boeing share tech for market access (e.g., turbines, 787 parts)

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China's Export Boom Comes at a Cost: How to Make Factories Play Fair

- Retailers place pressure on Chinese suppliers to produce cheap goods at the lowest possible costs
- Demand for cheap goods creates issues
  - Product safety
  - Quality of the environment
  - Labor protections

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India: Breaking Out of the Low-Income Classification

- Agriculture, handicrafts, manufacturing, services
- Two-thirds of Indian workforce in agriculture
- Services and technical support–young, English-fluent Indians are increasingly involved
- Improvement in infrastructure
  - Roads, electric power generation, rail freight, and ports

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India: Breaking Out of the Low-Income Classification

- High rate of population growth
  - Major advantage in almost limitless labor supply and consumer demand
  - Necessity of investing in education and health care
  - Creating adequate opportunities for employment

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India: Breaking Out of the Low-Income Classification

- Systematically deregulate sectors
  - Retailing, news media, and banking
- Eliminate preferences for small-scale, inefficient producers
- Reform agriculture to generate jobs in rural areas

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